

Real Change or a Lookalike? A Surrogate-Substitution Audit of Two-Stage Latent Diffusion Synthesis for Brain-Metastasis MRI

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Code & checkpoints: github.com/jgn2pyh2wn-afk/brain_tumour_predict

1. Motivation

Does a generative medical-imaging result really come from the generator? Contour and texture metrics are routinely reported as evidence for 3D brain-tumour synthesis, yet they are rarely controlled for the pipeline wrapped around the sampler. We build a self-contained two-stage latent diffusion model (LDM) for 64³ T1-Gd brain-metastasis volumes and audit it by surrogate substitution: the sampler is replaced, one surrogate at a time, under an identical post-process. The protocol runs in seconds on CPU and is written to be transferred to other pipelines; brain-metastasis follow-up is the setting that motivates the test, not a clinical claim.

2. Method

A 3D convolutional autoencoder maps a 64×64×64 T1-Gd volume to a 4×16×16×16 latent; a Diffusion Transformer (dim 64, 2 blocks, 4 heads, patch 2×2×2, 512 tokens; $T = 200$, linear $\beta \ 1 \times 10^{-4} \rightarrow 0.02$) predicts latent noise conditioned on a globally pooled latent. Stage 1 selects the starting noise level from the prescribed step span (AdaNI); stage 2 re-noises to 0.15 T and refines texture. A structure-preserving post-process confines the residual to a lesion-candidate mask and clips it at $c = 0.08$ with strength $s = 0.35$; all components are custom-trained.

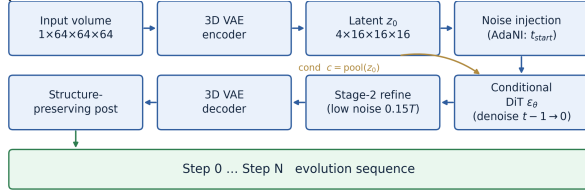


Fig. 1. Two-stage latent-diffusion framework (schematic); every component is custom-trained.

3. Non-uniform time weighting

Endpoint weights follow the normalised volume progress of an explicit growth law (exponential / Gompertz / logistic-Fisher-KPP) with an equivalent volume-doubling time T_d replacing uniform step/ N interpolation. The law is a monotone re-parameterisation of the step axis, not a cohort-calibrated growth model (T_d is not fitted here); what it buys is that the sequence is bounded by construction, since the post-process clips the model’s own contribution to $|\Delta_{\text{gen}}| \leq s \cdot c = 0.028$.

4. Results: structure preservation (Fig. 2)

On $N = 100$ steps over 51 cases: soft contour IoU 0.9836 ± 0.0047 , hard IoU 0.9881 ± 0.0035 , texture correlation 0.9808 ± 0.0012 , **candidate-region** mean change -0.0219 ± 0.0023 — the region is the heuristic candidate field (24.4% of brain), **not** the lesion; outside/inside ratio 0.633 (0.416–0.863); pre-clip reconstruction PSNR 15.993 ± 1.497 dB.

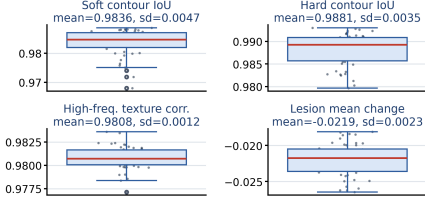


Fig. 2. Progression metrics over the 51-case test set ($N=100$ steps; box=IQR, dots=per-case).

5. Expert-lesion cross-check ($n = 105$; calibration audit)

Expert *seg* labels exist only in the publisher’s training split — the 105 studies this model trained on — so this is a mask-calibration audit, not held-out performance. The candidate field is 125× the lesion volume, covering 29.0% of lesion voxels, and only 0.37% of it is lesion. The same statistics inside expert masks (trilinear / max-pool bound): mean change -0.0494 ± 0.0237 / -0.0239 ± 0.0090 (candidate field, same 105: -0.0214 ± 0.0025); texture correlation 0.9952 ± 0.0031 / 0.9933 ± 0.0047 (0.9900 ± 0.0013); outside/inside ratio 0.509 ± 0.407 / 0.772 ± 0.243 (0.653 ± 0.100). Magnitudes move up to 2.3×; the verdict does not.

6. Qualitative progression (Fig. 3)

Three cases, Step 0→100, through the code’s own display chain (per-slice percentile stretch; stored data unchanged). Black marks voxels where $m \times w(t) > 0.35$, m the heuristic lesion-candidate field: a **prescribed schedule overlay — not predicted growth** — on the axial slice with the largest interior mask support ($Z = 26/27/38$); it is **not** a lesion segmentation (§5).

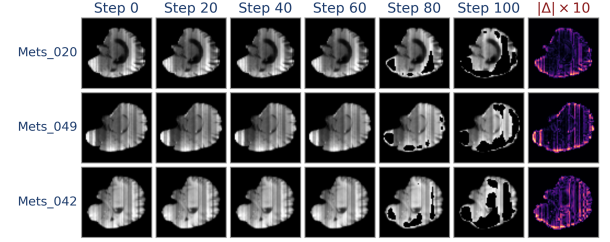


Fig. 3. Three cases, Step 0→100. Black = the prescribed overlay of §6, not predicted growth. Right: 10× the model’s own change (raw intensity; saturates at 0.1).

7. Generation-isolation audit (Fig. 4)

Replacing the sampler by frozen sampling, a constant latent or another patient’s result leaves **all four metrics bit-identical**; a source copy and pure noise land in the same band (per-case ranges 0.0034 / 0.0048 / 0.0188 / 0.0129). The collapse survives the region fix of §5: inside expert lesions on the 105 labelled studies, the true sampler, a second seed, frozen sampling, another patient’s output and a constant latent give **identical** mean change (-0.0494 ± 0.0237) and absolute change (0.05206 ± 0.02115); deleting generation moves the first by only 16%. Root cause: the decoder’s final Sigmoid saturates (latent magnitude 11.43; 99.97% of candidate-core voxels exceed the 0.08 clip), so these numbers characterise the post-processing chain, not the sampler.

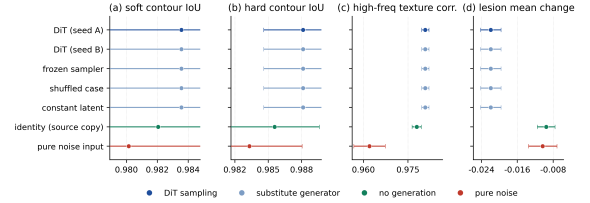


Fig. 4. Replacing the sampler with surrogates leaves the reported metrics unchanged.

8. Root cause is repairable — but not sufficient

Two changes only (EMA latent standardisation; Hardtanh(0,1) decoder output) under an otherwise identical recipe (4.5 min) give PSNR $15.993 \pm 1.497 \rightarrow 29.299 \pm 1.318$ dB, decoder std $0.0010 \rightarrow 0.1447$, core clipping saturation $99.97\% \rightarrow 34.98\%$, surrogate difference $0.000 \rightarrow 0.0433$ — a non-degenerate decoder, but still no lesion-specific generation (core/ring contrast 0.990 vs. 1.412).

9. Take-home and limits

Contribution. A self-contained two-stage LDM released with code, weights and scripts; a transferable surrogate-substitution audit (which reported numbers survive sampler replacement?); expert-lesion calibration of a heuristic mask; and a clip-saturation failure mode with a verified fix. **Limits.** One pipeline only; no longitudinal ground truth (the step index is a generation condition, not clinical time); T_d is uncalibrated; the expert-lesion audit uses training studies, because labels exist only there.

Compliance with Ethical Standards

Data and ethics. BrainMetShare [5], 156 studies, publisher’s split (105/51), non-commercial research licence; a retrospective, de-identified secondary analysis, so no new data were collected and no ethics approval or consent was required. **Conflicts of interest / funding:** none. **Reproducibility.** Code, checkpoints and scripts are released; a paper-to-code checker re-derives every quantitative claim of the accompanying full paper (245 pass / 0 fail). **AI disclosure.** The AI system OpenAI Codex was used for (i) *code assistance* — implementing and refactoring the training, inference, audit and figure scripts — and (ii) *paper layout and typesetting*. Every number is emitted by the released code; the author directed, reviewed and edited all content.

References

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